COMPUTER ENGINEERING

QUEZON CITY

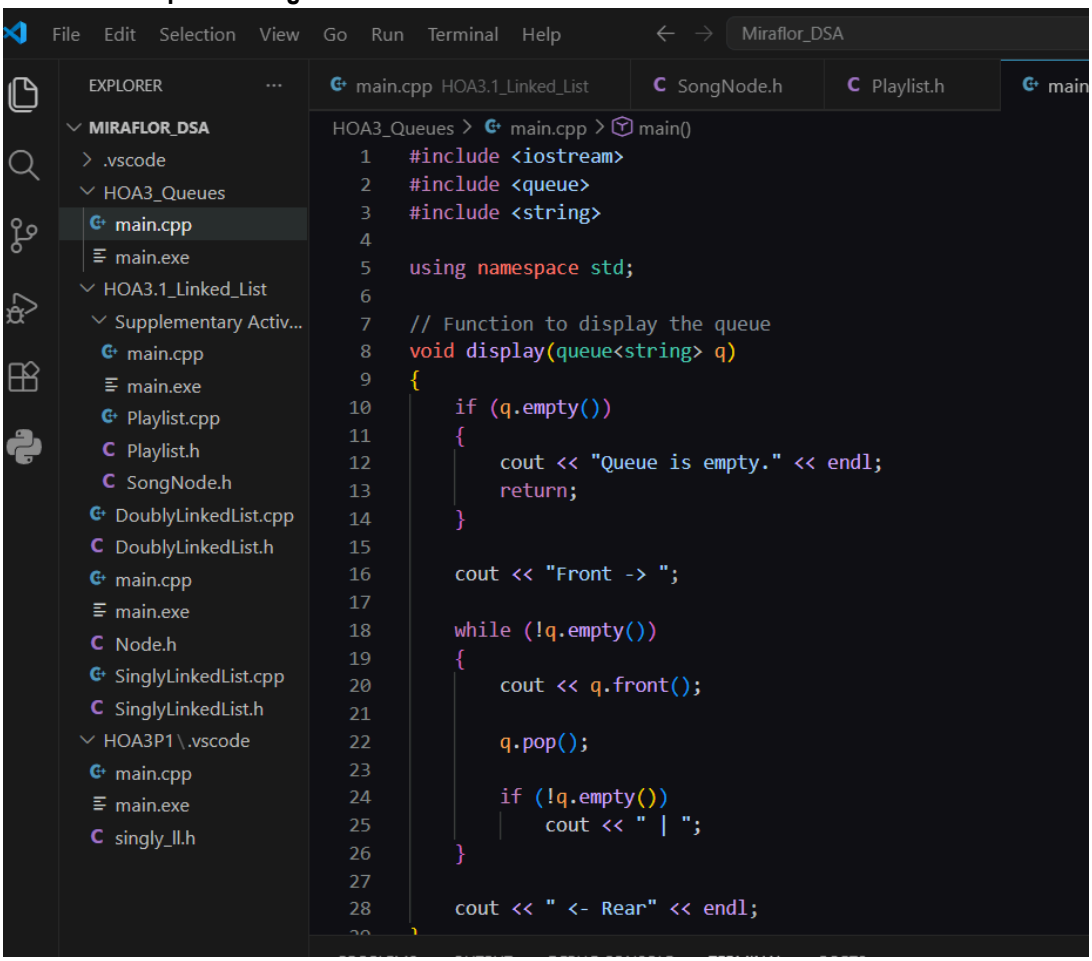
COURSE CODE/TITLE:	**CPE 010**	**SECTION:**	**CPE21S4**
DATE PERFORMED:	**07/05/2026**	**DATE SUBMITTED:**	**07/05/2026**
NAME:	**Seth Carlo T. Miraflor**	**INSTRUCTOR:**	**Engr. Jimlord Quejado**

ACTIVITY NO. 5.1 HANDS-ON ACTIVITY QUEUES

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A: Create queue using C++ STL



HOA3_Queues > main.cpp > main()

```cpp
int main()
{
    queue<string> students;

    // Array of students' names
    string names[] = {
        "SETH",
        "EMMAN",
        "TRISHA",
        "JAMES",
        "ANGEL"
    };

    cout << "===== CREATE QUEUE =====" << endl;
    display(students);

    cout << "\n===== ENQUEUE OPERATIONS =====" << endl;

    for(int i = 0; i < 5; i++)
    {
        cout << "\nEnqueue: " << names[i] << endl;
        students.push(names[i]);

        display(students);

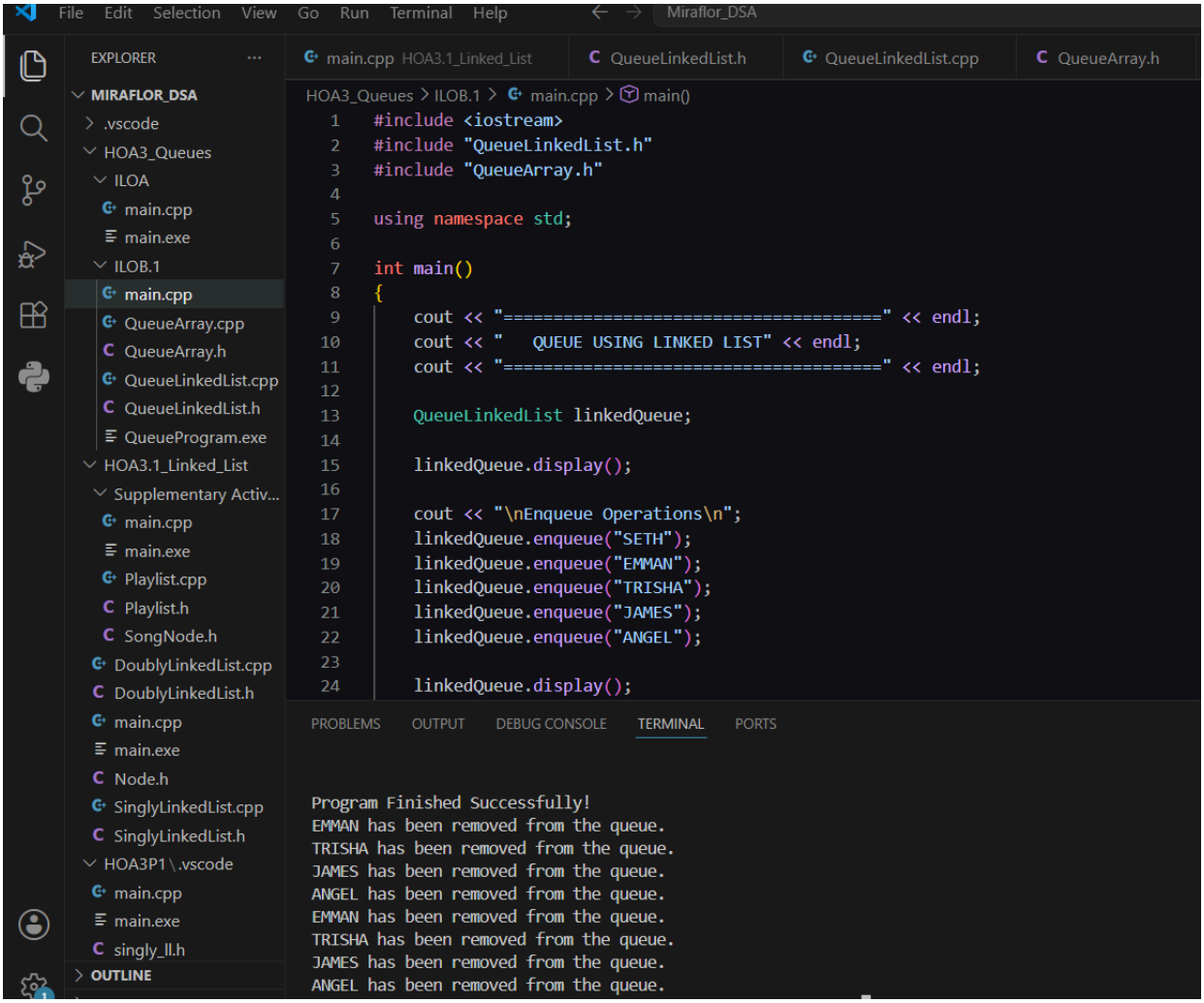
        cout << "Front: " << students.front() << endl;
        cout << "Back : " << students.back() << endl;
        cout << "Size : " << students.size() << endl;
```

HOA3_Queues > main.cpp > main()

```
61      cout << "\n===== DEQUEUE OPERATION =====" << endl;
62
63      cout << "Removed: " << students.front() << endl;
64      students.pop();
65
66      display(students);
67
68      cout << "\n===== FRONT ELEMENT =====" << endl;
69      cout << students.front() << endl;
70
71      cout << "\n===== BACK ELEMENT =====" << endl;
72      cout << students.back() << endl;
73
74      cout << "\n===== IS EMPTY? =====" << endl;
75
76      if(students.empty())
77          cout << "Yes" << endl;
78      else
79          cout << "No" << endl;
80
81      cout << "\n===== DESTROY QUEUE =====" << endl;
82
83      while(!students.empty())
84      {
85          students.pop();
86      }
87
88      display(students);
```

```
    return 0;
}
```

B.1. Linked List Implementation


File Edit Selection View Go Run Terminal Help
Miraflor_DSA
EXPLORER
MIRAFLOR_DSA
.vscode
HOA3_Queues
ILOA
main.cpp
main.exe
ILOB.1
main.cpp
QueueArray.cpp
QueueArray.h
QueueLinkedList.cpp
QueueLinkedList.h
QueueProgram.exe
HOA3.1_Linked_List
Supplementary Activ...
main.cpp
main.exe
Playlist.cpp
Playlist.h
SongNode.h
DoublyLinkedList.cpp
DoublyLinkedList.h
main.cpp
main.exe
Node.h
SinglyLinkedList.cpp
SinglyLinkedList.h
HOA3P1 \ .vscode
main.cpp
main.exe
singly_ll.h
OUTLINE
main.cpp HOA3.1_Linked_List
QueueLinkedList.h
QueueLinkedList.cpp
QueueArray.h
HOA3_Queues > ILOB.1 > main.cpp > main()
1 #include <iostream>
2 #include "QueueLinkedList.h"
3 #include "QueueArray.h"
4
5 using namespace std;
6
7 int main()
8 {
9 cout << "====================================" << endl;
10 cout << " QUEUE USING LINKED LIST" << endl;
11 cout << "====================================" << endl;
12
13 QueueLinkedList linkedQueue;
14
15 linkedQueue.display();
16
17 cout << "\nEnqueue Operations\n";
18 linkedQueue.enqueue("SETH");
19 linkedQueue.enqueue("EMMAN");
20 linkedQueue.enqueue("TRISHA");
21 linkedQueue.enqueue("JAMES");
22 linkedQueue.enqueue("ANGEL");
23
24 linkedQueue.display();
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Program Finished Successfully!
EMMAN has been removed from the queue.
TRISHA has been removed from the queue.
JAMES has been removed from the queue.
ANGEL has been removed from the queue.
EMMAN has been removed from the queue.
TRISHA has been removed from the queue.
JAMES has been removed from the queue.
ANGEL has been removed from the queue.

```
====================================
   QUEUE USING LINKED LIST
====================================
Queue is empty.

Enqueue Operations
SETH has been added to the queue.
EMMAN has been added to the queue.
TRISHA has been added to the queue.
JAMES has been added to the queue.
ANGEL has been added to the queue.
Front -> SETH -> EMMAN -> TRISHA -> JAMES -> ANGEL <- Back

Front: SETH
Back : ANGEL
Size : 5

Dequeue Operation
SETH has been removed from the queue.
Front -> EMMAN -> TRISHA -> JAMES -> ANGEL <- Back

Front: EMMAN
Back : ANGEL
Size : 4

Testing Copy Constructor
EMMAN has been added to the queue.
TRISHA has been added to the queue.
JAMES has been added to the queue.
ANGEL has been added to the queue.
Front -> EMMAN -> TRISHA -> JAMES -> ANGEL <- Back

Testing Copy Assignment Operator
EMMAN has been added to the queue.
TRISHA has been added to the queue.
JAMES has been added to the queue.
ANGEL has been added to the queue.
```

```
Front -> EMMAN -> TRISHA -> JAMES -> ANGEL <- Back

Clearing Queue...
EMMAN has been removed from the queue.
TRISHA has been removed from the queue.
JAMES has been removed from the queue.
ANGEL has been removed from the queue.
Queue is empty.
```

```
=====================================
      QUEUE USING ARRAY
=====================================
Queue is empty.

Enqueue Operations
SETH has been added to the queue.
EMMAN has been added to the queue.
TRISHA has been added to the queue.
JAMES has been added to the queue.
ANGEL has been added to the queue.
Front -> SETH -> EMMAN -> TRISHA -> JAMES -> ANGEL <- Back

Front: SETH
Back : ANGEL
Size : 5

Dequeue Operation
SETH has been removed from the queue.
Front -> EMMAN -> TRISHA -> JAMES -> ANGEL <- Back

Front: EMMAN
Back : ANGEL
Size : 4

Testing Copy Constructor
Front -> EMMAN -> TRISHA -> JAMES -> ANGEL <- Back

Testing Copy Assignment Operator
Front -> EMMAN -> TRISHA -> JAMES -> ANGEL <- Back

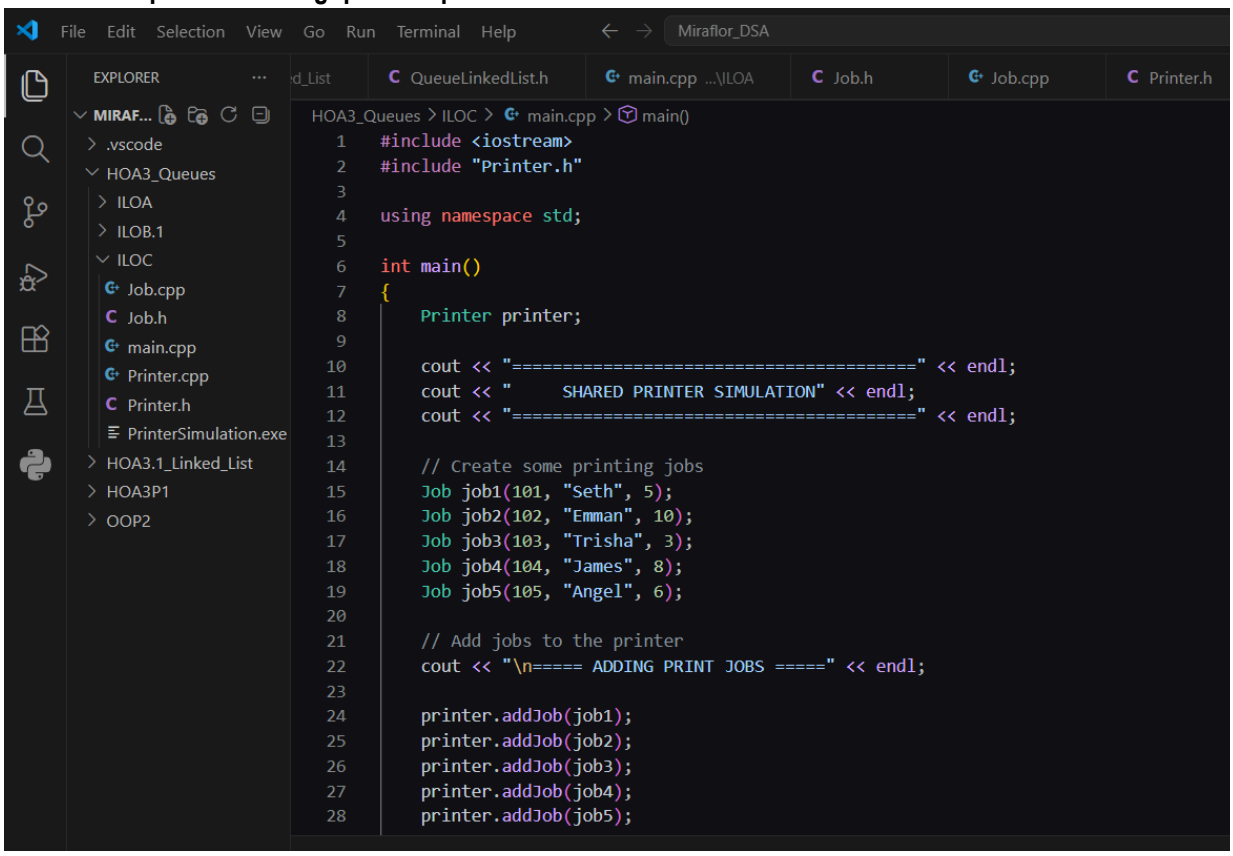
Clearing Queue...
Queue is empty.
```

```
Program Finished Successfully!
EMMAN has been removed from the queue.
TRISHA has been removed from the queue.
JAMES has been removed from the queue.
ANGEL has been removed from the queue.
EMMAN has been removed from the queue.
TRISHA has been removed from the queue.
JAMES has been removed from the queue.
ANGEL has been removed from the queue.
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

ILO C: Solve problems using queue implementation



```
========================================
     SHARED PRINTER SIMULATION
========================================

===== ADDING PRINT JOBS =====

New print job added!
Job ID: 101
User: Seth
Pages: 5

New print job added!
Job ID: 102
User: Emman
Pages: 10

New print job added!
Job ID: 103
User: Trisha
Pages: 3

New print job added!
Job ID: 104
User: James
Pages: 8

New print job added!
Job ID: 105
User: Angel
Pages: 6

===== PENDING PRINT JOBS =====

Job ID: 101
User: Seth
Pages: 5

Job ID: 102
User: Emman
Pages: 10
```

```
Job ID: 103
User: Trisha
Pages: 3

Job ID: 104
User: James
Pages: 8

Job ID: 105
User: Angel
Pages: 6

Total pending jobs: 5

===== PROCESSING FIRST JOB =====

Processing print job...
Job ID: 101
User: Seth
Pages: 5
Printing completed.

===== PENDING PRINT JOBS =====

Job ID: 102
User: Emman
Pages: 10

Job ID: 103
User: Trisha
Pages: 3

Job ID: 104
User: James
Pages: 8

Job ID: 105
User: Angel
Pages: 6
```

```
Total pending jobs: 4

===== PROCESSING REMAINING JOBS =====

===== PROCESSING ALL JOBS =====

Processing print job...
Job ID: 102
User: Emman
Pages: 10
Printing completed.

Processing print job...
Job ID: 103
User: Trisha
Pages: 3
Printing completed.

Processing print job...
Job ID: 104
User: James
Pages: 8
Printing completed.

Processing print job...
Job ID: 105
User: Angel
Pages: 6
Printing completed.

All print jobs have been processed.

===== CHECKING PRINTER QUEUE =====
The printer queue is empty.

========================================
       SIMULATION FINISHED
========================================
PS C:\Users\Admin\Desktop\Miraflor_DSA\HOA3_Queues\ILOC>
```

COMPUTER ENGINEERING

QUEZON CITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Chatgpt**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Idea generation**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used ai for an idea that how im gonna start with off and combine it with my idea of how im gonna make it work**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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